

ME101: (3 1 0 8)

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# 3D KINEMATICS OF RIGID BODIES

# 3-D Kinematics of Rigid Bodies

Several possible motions

## Translation

### :: Rectilinear Translation

- Any two points in the body will move along parallel straight lines

### :: Curvilinear Translation

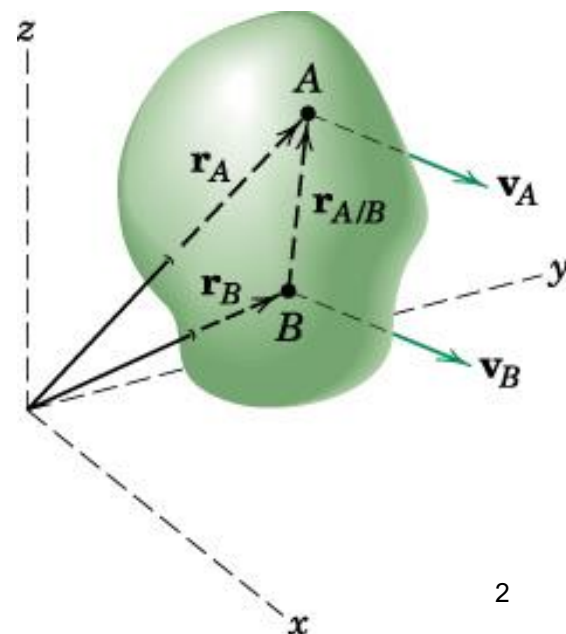
- Any two points in the body will move along congruent curves

:: Line joining the two points will always remain parallel to its original position

$$\mathbf{r}_A = \mathbf{r}_B + \mathbf{r}_{A/B} \quad \mathbf{v}_A = \mathbf{v}_B \quad \mathbf{a}_A = \mathbf{a}_B$$

$\mathbf{r}_{A/B}$  is constant and its time derivative is zero

:: All points in the body have same velocity and same acceleration



# 3-D Kinematics of Rigid Bodies

## Fixed Axis Rotation

::  $\omega$  does not change its direction since it lies along the fixed axis

:: Any point  $A$  in the body (not on the axis) moves in a circular arc in a plane normal to the axis; its velocity:

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} \quad \text{since } \mathbf{r} = \mathbf{h} + \mathbf{b} \text{ and } \boldsymbol{\omega} \times \mathbf{h} = \mathbf{0}$$

And acceleration of  $A$  is given by time derivative:

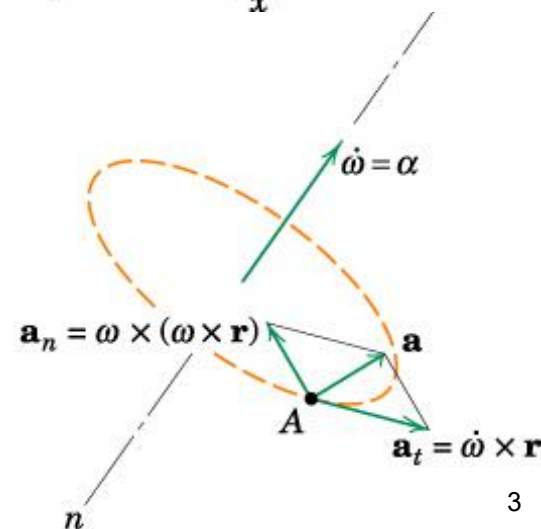
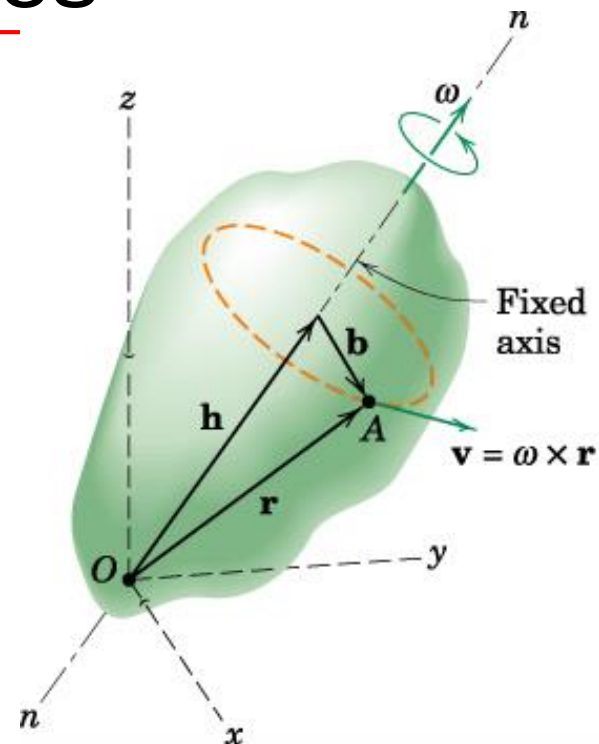
$$\mathbf{a} = \dot{\boldsymbol{\omega}} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) \quad \text{since } \mathbf{r}' = \mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$$

The normal and tangential components of  $\mathbf{a}$  for the circular motion:

$$a_n = |\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})| = b\omega^2 \quad a_t = |\dot{\boldsymbol{\omega}} \times \mathbf{r}| = b\alpha$$

Since  $\mathbf{v}$  and  $\mathbf{a}$  are perpendicular to  $\boldsymbol{\omega}$  and  $\dot{\boldsymbol{\omega}}$ :

$$\mathbf{v} \cdot \boldsymbol{\omega} = 0, \mathbf{v} \cdot \dot{\boldsymbol{\omega}} = 0, \mathbf{a} \cdot \boldsymbol{\omega} = 0, \text{ and } \mathbf{a} \cdot \dot{\boldsymbol{\omega}} = 0$$



# 3-D Kinematics of Rigid Bodies

## Rotation about a Fixed Point

:: Change in direction of angular velocity vector  $\rightarrow$  more general rotation

## Proper Vectors

:: Rotation vectors are not obeying the parallelogram law of vector addition  
 $\rightarrow$  not treated as proper vectors

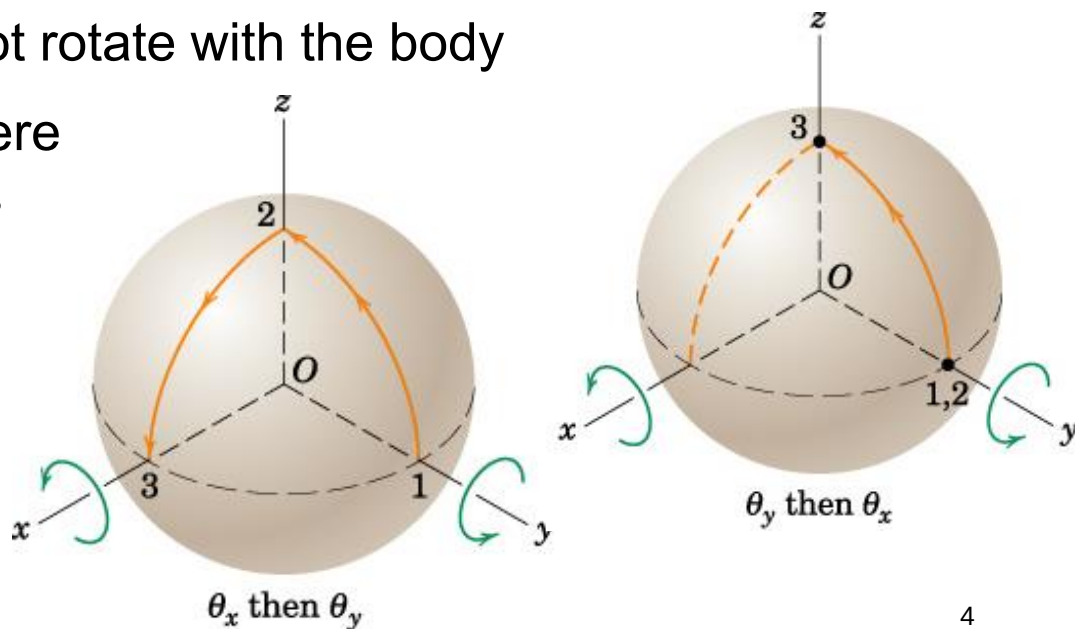
Consider a solid sphere cut from a rigid body confined to rotate @ fixed point  $O$

:: x-y-z axes are fixed and do not rotate with the body

:: Consider  $90^\circ$  rotations of sphere

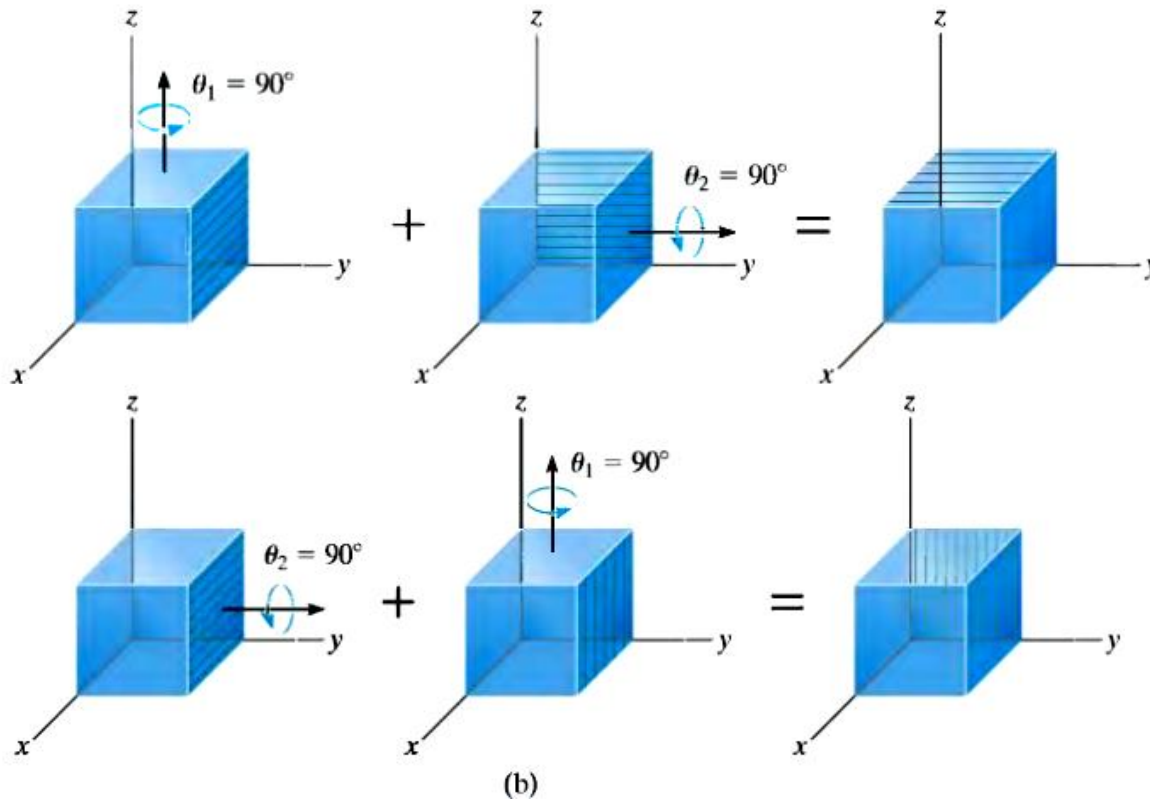
@ x and y axes in two cases

1. @ x- then @ y-axis
2. @ y- then @ x-axis



# 3-D Kinematics of Rigid Bodies

- Rotation about a Fixed Point



$$\theta_1 + \theta_2 \neq \theta_2 + \theta_1$$

- $\therefore$  The two cases do not give the same final position
- $\therefore$  Finite rotations do not generally obey the parallelogram law of vector addition  $\rightarrow$  not commutative
- $\rightarrow$  Finite rotations may not be treated as proper vectors

# 3-D Kinematics

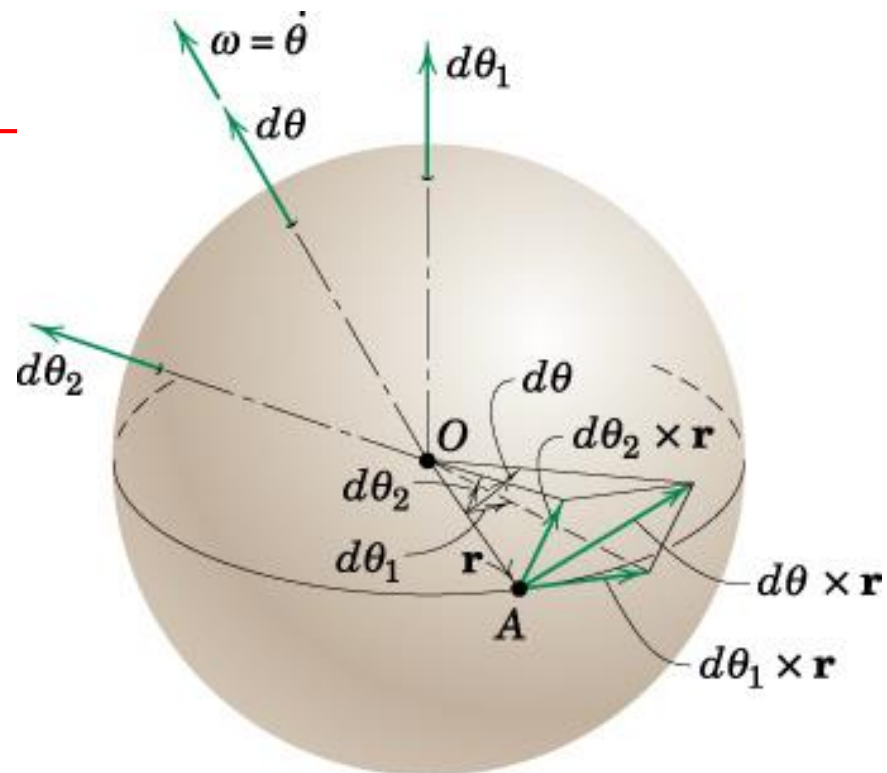
## Rotation about a Fixed Point

### Infinitesimal Rotation

Combined effect of two infinitesimal rotations  $d\theta_1$  and  $d\theta_2$  of a rigid body @ respective axes through fixed point  $O$

Displacement of  $A$  due to  $d\theta_1$ :  $d\theta_1 \times \mathbf{r}$

Displacement of  $A$  due to  $d\theta_2$ :  $d\theta_2 \times \mathbf{r}$



Same resultant displacement due to either order of addition of these infinitesimal displacements:

$$\rightarrow d\theta_1 \times \mathbf{r} + d\theta_2 \times \mathbf{r} = (d\theta_1 + d\theta_2) \times \mathbf{r} = d\theta \times \mathbf{r}$$

$\rightarrow$  Two rotations equivalent to a single rotation  $d\theta = d\theta_1 + d\theta_2$

$\rightarrow$  Addition of angular velocities  $\omega_1 = \dot{\theta}_1$  and  $\omega_2 = \dot{\theta}_2$  gives:  $\omega = \dot{\theta} = \omega_1 + \omega_2$

$\rightarrow$  Infinitesimal rotations obey the parallelogram law of vector addition

$\rightarrow$  At **any instant of time**, a body with **one fixed point** rotates instantaneously @ a **particular axis** passing through the **fixed point**

# 3-D Kinematics of Rigid Bodies

## Rotation about a Fixed Point

### Instantaneous Axis of Rotation

:: Rotation of solid rotor with black particles embedded on its surface

::  $\omega_1$  and  $\omega_2$  are steady angular velocities

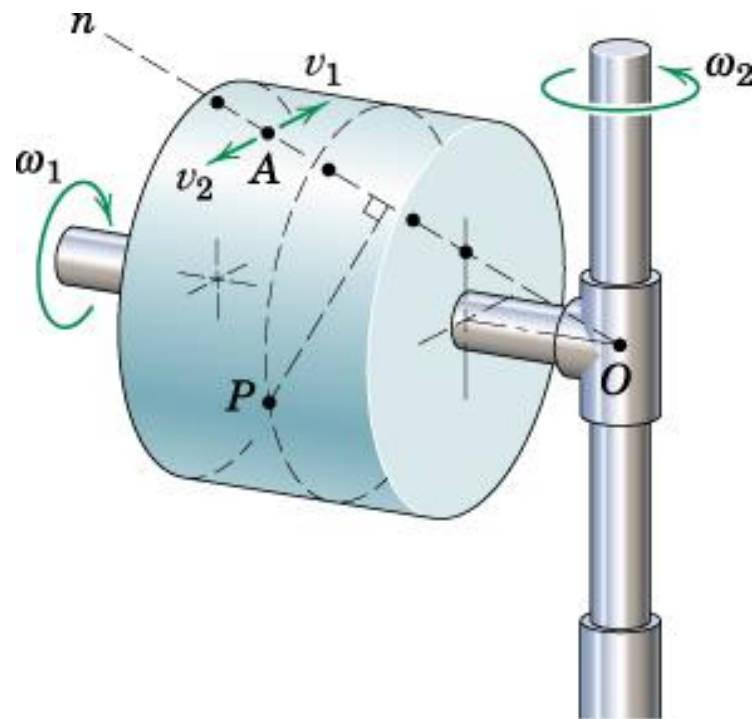
:: Along line  $O-n$ , velocity of particles will be momentarily zero (the line will be sharply defined)

:: The line of points with no velocity

→ Instantaneous position of the Axis of Rotation

:: Any point on this line, e.g.,  $A$ , would have equal and opposite velocity components  $v_1$  and  $v_2$  due to  $\omega_1$  and  $\omega_2$ , respectively

All particles of the body, except those on line  $O-n$ , rotate momentarily in circular arcs @ the instantaneous axis of rotation



# 3-D Kinematics

## Rotation about a Fixed Point

### Instantaneous Axis of Rotation

:: Change of position of rotation axis both in space and relative to the body

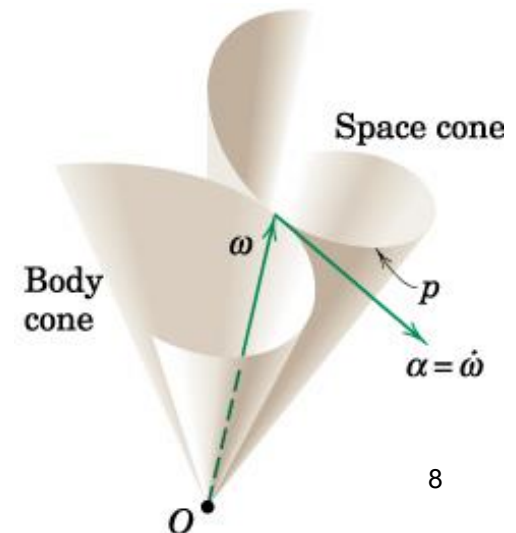
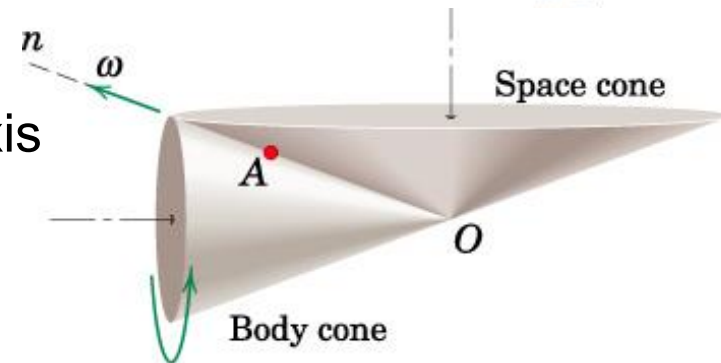
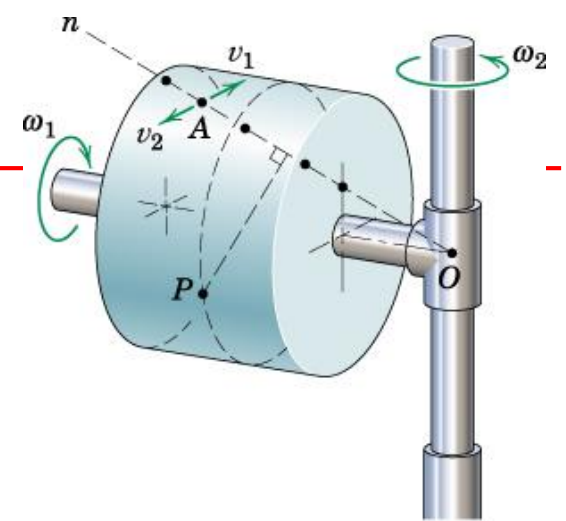
### Body and Space Cones

:: Relative to the cylinder, the instantaneous axis of rotation  $O-A-n$  generates a right circular cone @ the cylinder axis  $O \rightarrow$  **Body Cone**

:: Instantaneous axis of rotn also generates a right circular cone @ the vertical axis  $\rightarrow$  **Space Cone**

:: Rolling of body cone on space cone

:: Angular vel  $\omega$  of the body located along the common element of the two cones





# 3-D Kinematics

## Rotation about a Fixed Point

### Angular Acceleration

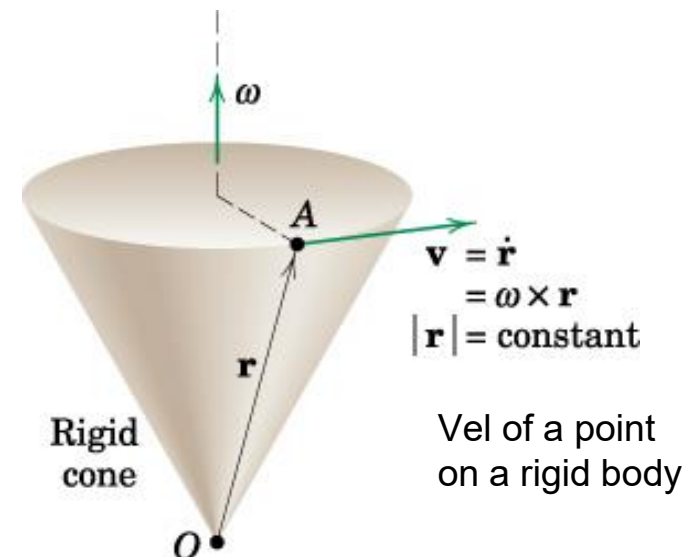
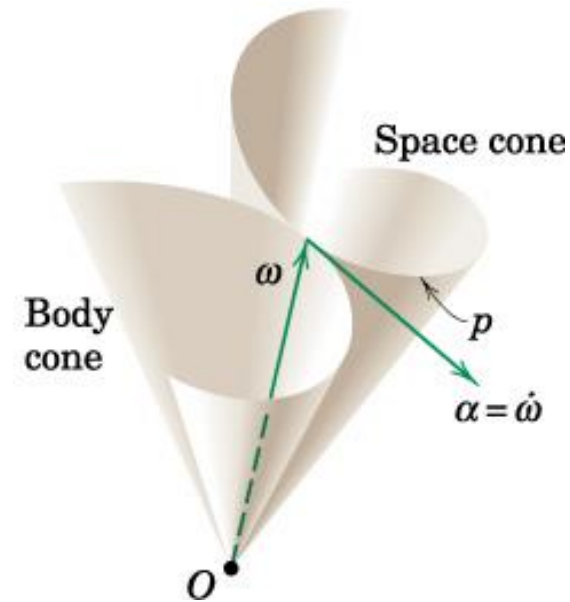
:: Planar motion

- scalar  $\alpha$  reflects the change in magnitude of  $\omega$

:: 3-D motion

- vector  $\alpha$  reflects the change in the direction and magnitude of  $\omega$

::  $\alpha$  - tangent vector to the space curve  $p$  in the direction of the change in  $\omega$



# 3-D Kinematics

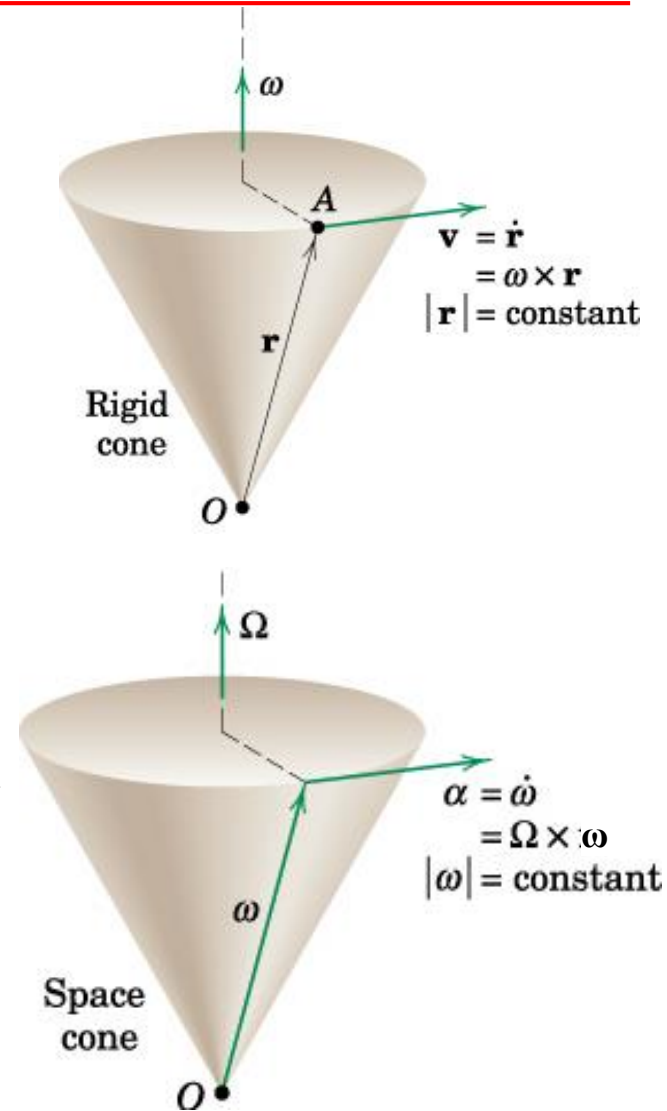
## Rotation about a Fixed Point

If the magnitude of  $\omega$  remains constant,  $\alpha$  is normal to  $\omega$

Let  $\Omega$  be the angular velocity with which the vector  $\omega$  itself rotates (*precesses*) as it forms the space cone

$$\rightarrow \alpha = \Omega \times \omega$$

$\therefore \alpha$ ,  $\Omega$ , and  $\omega$  have the same relationship as that of  $\mathbf{v}$ ,  $\omega$ , and  $\mathbf{r}$  defining the vel of a point A in a rigid body



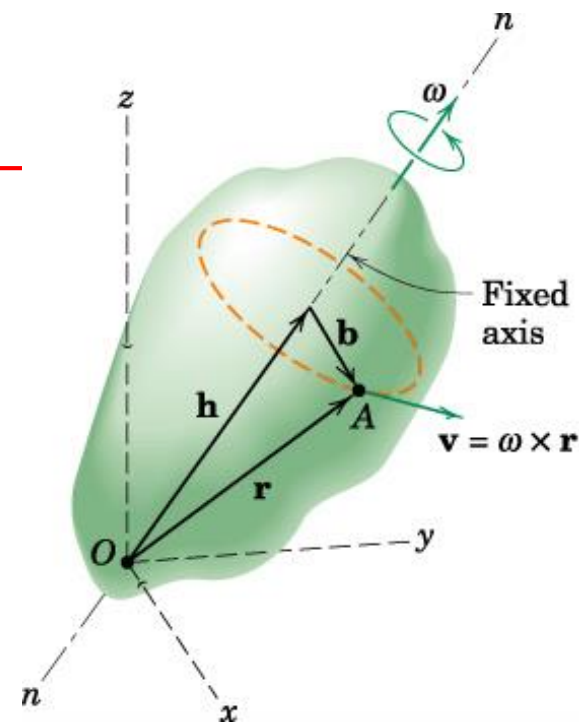
# 3-D Kinematics

## Rotation about a Fixed Point

When a rigid body rotates @ a fixed point  $O$  with the instantaneous axis of rotation  $n$ - $n$ , vel  $\mathbf{v}$  and accln  $\mathbf{a} = \dot{\mathbf{v}}$  of any point  $A$  in the body are given by the same expressions derived when the axis was fixed:

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$$

$$\mathbf{a} = \dot{\boldsymbol{\omega}} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$



## Rotation about a Fixed Axis vs Rotation about a Fixed Point

For rotation @ a fixed point:

- angular accln will have a comp normal to  $\boldsymbol{\omega}$  due to change in dirn of  $\boldsymbol{\omega}$ , as well as a comp in the dirn of  $\boldsymbol{\omega}$  to reflect any change in the magnitude of  $\boldsymbol{\omega}$
- Although any point on the rotation axis  $n$ - $n$  momentarily will have zero velocity, it will not have zero accln as long as  $\boldsymbol{\omega}$  is changing its direction

For rotation @ a fixed axis:

- angular accln will have only one component along the fixed axis to reflect any change in the magnitude of  $\boldsymbol{\omega}$ .
- Points which lie on the fixed axis of rotation will have no velocity or accln

# Example (1) on 3-D Kinematics

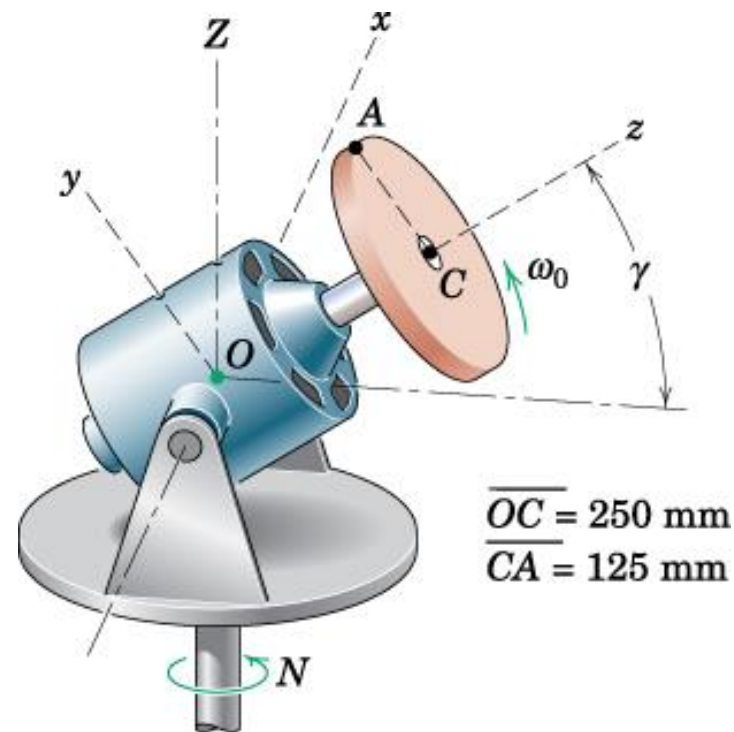
The electric motor with an attached disk is running at a constant low speed of 120 rev/min in the direction shown. Its housing and mounting base are initially at rest. The entire assembly is next set in rotation about the vertical Z-axis at the constant rate  $N = 60$  rev/min with a fixed angle  $\gamma$  of  $30^\circ$ . Determine (a) the angular velocity and angular acceleration of the disk, (b) the space and body cones, and (c) the velocity and acceleration of point A at the top of the disk for the instant shown.

## Solution

Axes  $x$ ,  $y$ ,  $z$  with unit vectors  $\mathbf{i}$ ,  $\mathbf{j}$ , and  $\mathbf{k}$  are attached to the motor frame.

- $x$ -axis coincides with the horz axis through O @ which the motor tilts.
- $z$ -axis coincides with the rotor axis
- $y$ -axis will be as shown normal to  $x$  and  $z$
- $Z$ -axis is vertical and has unit vectors:

$$\mathbf{K} = \mathbf{j}\cos\gamma + \mathbf{k}\sin\gamma$$



# Example (1) on 3-D Kinematics

**Solution:**  $\mathbf{K} = \mathbf{j} \cos \gamma + \mathbf{k} \sin \gamma$

(a) The rotor and disk have two comp of angular vel:

@ z-axis:  $\omega_0 = 120(2\pi)/60 = 4\pi \text{ rad/s}$

@ Z-axis:  $\Omega = 60(2\pi)/60 = 2\pi \text{ rad/s}$

Angular velocity of the disk:

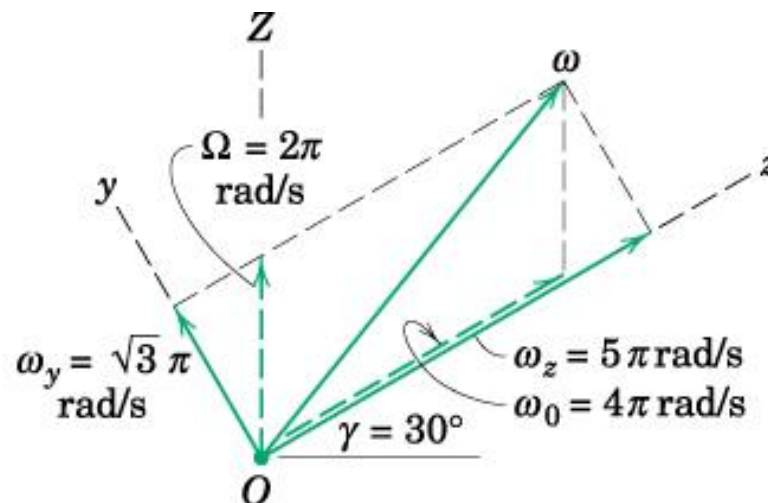
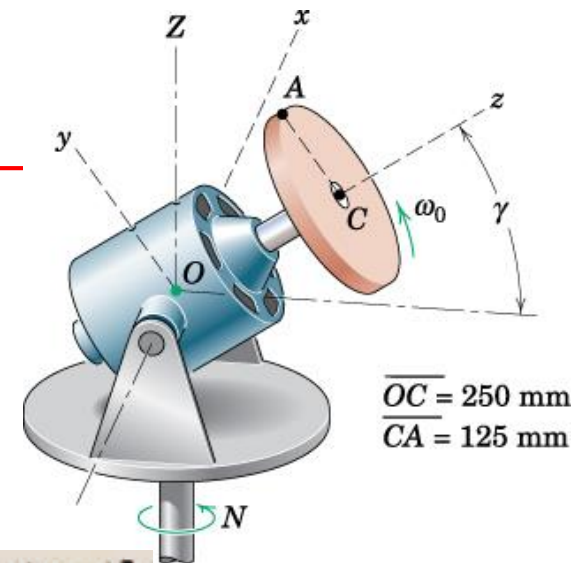
$$\boldsymbol{\omega} = \omega_0 + \boldsymbol{\Omega} = \omega_0 \mathbf{k} + \boldsymbol{\Omega} \mathbf{K}$$

$$= \omega_0 \mathbf{k} + \Omega(\mathbf{j} \cos \gamma + \mathbf{k} \sin \gamma) = (\Omega \cos \gamma) \mathbf{j} + (\omega_0 + \Omega \sin \gamma) \mathbf{k}$$

$$= (2\pi \cos 30^\circ) \mathbf{j} + (4\pi + 2\pi \sin 30^\circ) \mathbf{k} = \pi(\sqrt{3} \mathbf{j} + 5.0 \mathbf{k}) \text{ rad/sec}$$

It is clear from vector diagram that

$$\boldsymbol{\omega} = \omega_0 + \boldsymbol{\Omega} = \omega_y + \omega_z$$





# Example (1) on 3-D Kinematics

- (a) Since the magnitude of the angular vel  $\omega$  is constant (**steady precession**), Angular acceleration of the disk:

$$\alpha = \dot{\omega} = \Omega \times \omega$$

$$= \Omega(\mathbf{j} \cos \gamma + \mathbf{k} \sin \gamma) \times [(\Omega \cos \gamma)\mathbf{j} + (\omega_0 + \Omega \sin \gamma)\mathbf{k}]$$

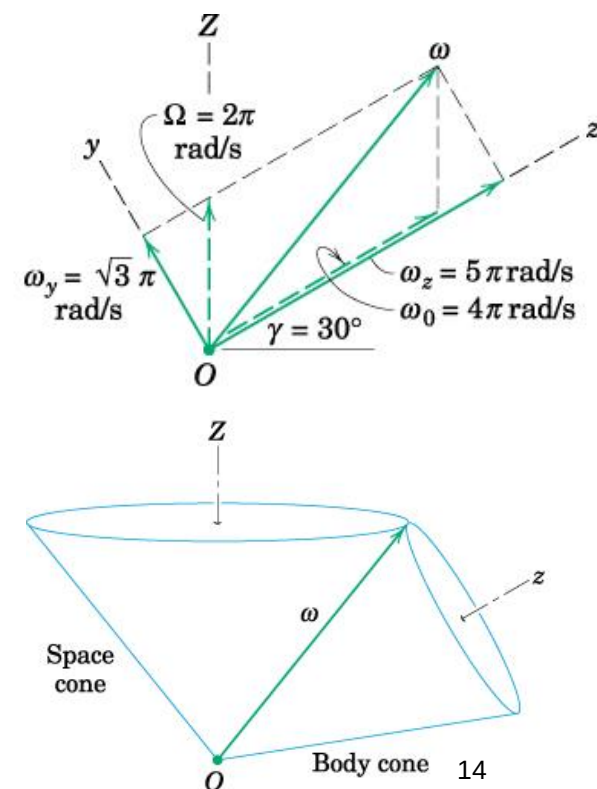
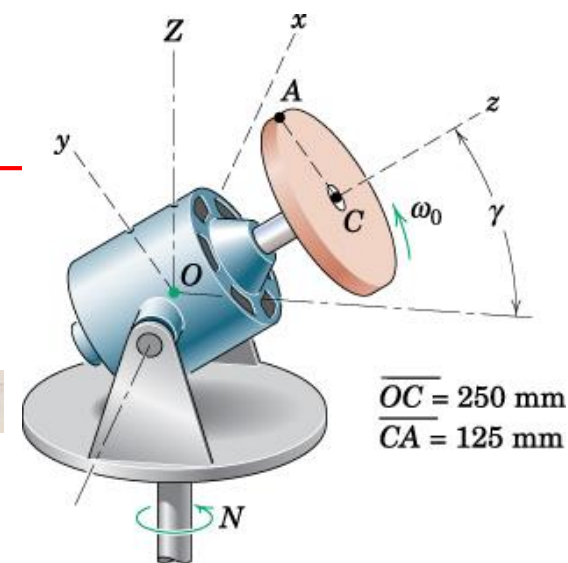
$$= \Omega(\omega_0 \cos \gamma + \Omega \sin \gamma \cos \gamma)\mathbf{i} - (\Omega^2 \sin \gamma \cos \gamma)\mathbf{i}$$

$$= (\Omega\omega_0 \cos \gamma)\mathbf{i} = \mathbf{i}(2\pi)(4\pi) \cos 30^\circ = 68.4\mathbf{i} \text{ rad/sec}^2$$

- (b) Space Cone and Body Cone:

Angular velocity vector  $\omega$  is the common element of the space and body cones.

Since magnitude of the angular vel  $\omega$  is constant, angular accln  $\alpha$  must be tangent to the base circle of the space cone and  $\alpha$  must be normal to  $\omega$   
 $\rightarrow \alpha$  must be along x-direction as calculated.



# Example (1) on 3-D Kinematics

**Solution:**  $\mathbf{K} = \mathbf{j}\cos\gamma + \mathbf{k}\sin\gamma$

(c) Linear instantaneous Vel and Accln of A

→ Rotation @ fixed axis can be considered because axis of rotation is fixed at the instant  $\gamma = 30^\circ$

→

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$$

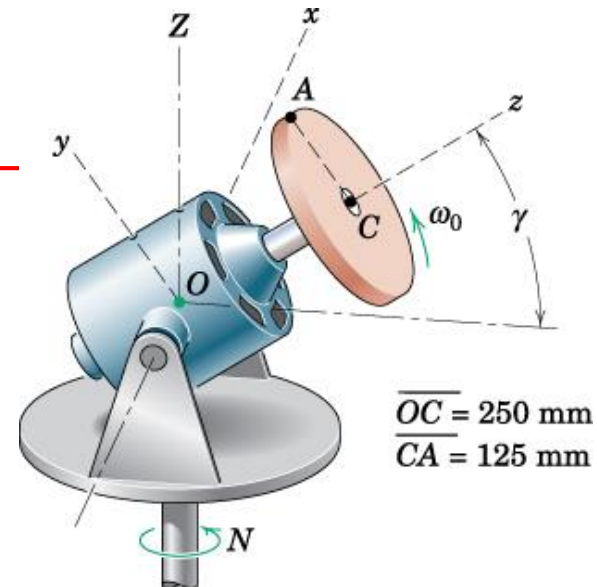
$$\mathbf{a} = \dot{\boldsymbol{\omega}} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$

Position vector of A for the instant considered:  $\mathbf{r} = 0.125\mathbf{j} + 0.250\mathbf{k}$  m

Velocity of point A =  $\boldsymbol{\omega} \times \mathbf{r}$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & \sqrt{3}\pi & 5\pi \\ 0 & 0.125 & 0.250 \end{vmatrix} = -0.6\mathbf{i} \text{ m/s}$$

Accln of point A:  $\mathbf{a} = \dot{\boldsymbol{\omega}} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) = \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times \mathbf{v}$   
 $= 68.4\mathbf{i} \times (0.125\mathbf{j} + 0.250\mathbf{k}) + \pi(\sqrt{3}\mathbf{j} + 5\mathbf{k}) \times (-0.6\mathbf{i})$   
 $\mathbf{a} = -26.6\mathbf{j} + 11.83\mathbf{k} \text{ m/s}^2$



# 3-D Kinematics

## General Motion in Three Dimension

:: Principles of relative motion (both translating & rotating reference axes)

### Translating Reference Axes

- $X$ - $Y$ - $Z$  :: fixed reference frame
- $x$ - $y$ - $z$  :: translating reference frame with any point  $B$  as the origin
- $\omega$  :: angular velocity of the rigid body

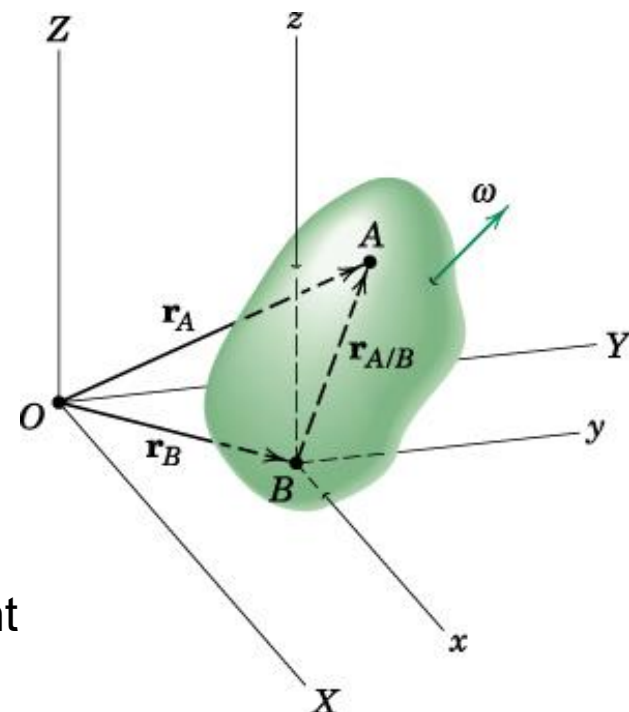
Linear velocity and accln of any point  $A$  in the body

$$\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{A/B}$$

-3 coplanar vectors in each eqn

$$\mathbf{a}_A = \mathbf{a}_B + \mathbf{a}_{A/B}$$

- Distance  $\overline{AB}$  should remain constant



- To an observer on  $x$ - $y$ - $z$ , the body will appear to rotate @  $B$
- Point  $A$  will appear to lie on a spherical surface with  $B$  as the center

→ General Motion  $\equiv$  translation of body with motion of  $B$   
+ rotation of body @  $B$



# 3-D Kinematics :: General motion

## Translating Reference Axes

$$\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{A/B}$$

$$\mathbf{a}_A = \mathbf{a}_B + \mathbf{a}_{A/B}$$

Relative motion terms represent effect of rotation about point  $B$

→ These are identical to the vel and accln relations obtained for a rigid body rotating @ a fixed point:

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$$

$$\mathbf{a} = \dot{\boldsymbol{\omega}} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$

Relative terms

→

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\omega} \times \mathbf{r}_{A/B}$$

$$\mathbf{a}_A = \mathbf{a}_B + \dot{\boldsymbol{\omega}} \times \mathbf{r}_{A/B} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{A/B})$$

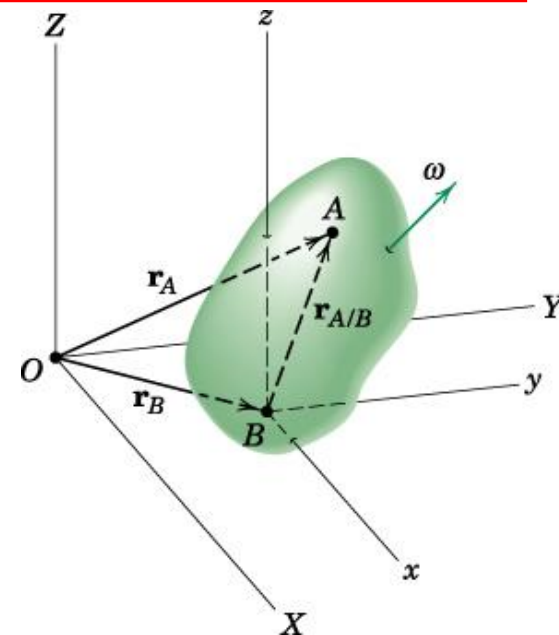
$\boldsymbol{\omega}$  and  $\dot{\boldsymbol{\omega}}$  :: instantaneous angular vel and angular accln of the body

If point  $A$  is chosen as a reference point then vel and accln relations for point  $B$ :

$\mathbf{r}_{B/A} = -\mathbf{r}_{A/B}$   $\boldsymbol{\omega}$  &  $\dot{\boldsymbol{\omega}}$  will be same for both cases

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$

$$\mathbf{a}_B = \mathbf{a}_A + \dot{\boldsymbol{\omega}} \times \mathbf{r}_{B/A} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{B/A})$$



# 3-D Kinematics

## Translating and Rotating Reference Axes

$$\dot{\mathbf{i}} = \boldsymbol{\omega} \times \mathbf{i} \quad \text{and} \quad \dot{\mathbf{j}} = \boldsymbol{\omega} \times \mathbf{j}$$

$$\dot{\mathbf{i}} = \boldsymbol{\Omega} \times \mathbf{i} \quad \dot{\mathbf{j}} = \boldsymbol{\Omega} \times \mathbf{j} \quad \dot{\mathbf{k}} = \boldsymbol{\Omega} \times \mathbf{k}$$

→ Time derivatives of the unit vectors attached to x-y-z

Vel and accln of point A:

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\omega} \times \mathbf{r} + \mathbf{v}_{\text{rel}}$$

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\Omega} \times \mathbf{r}_{A/B} + \mathbf{v}_{\text{rel}}$$

$$\mathbf{a}_A = \mathbf{a}_B + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{A/B} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{A/B}) + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \mathbf{a}_{\text{rel}}$$

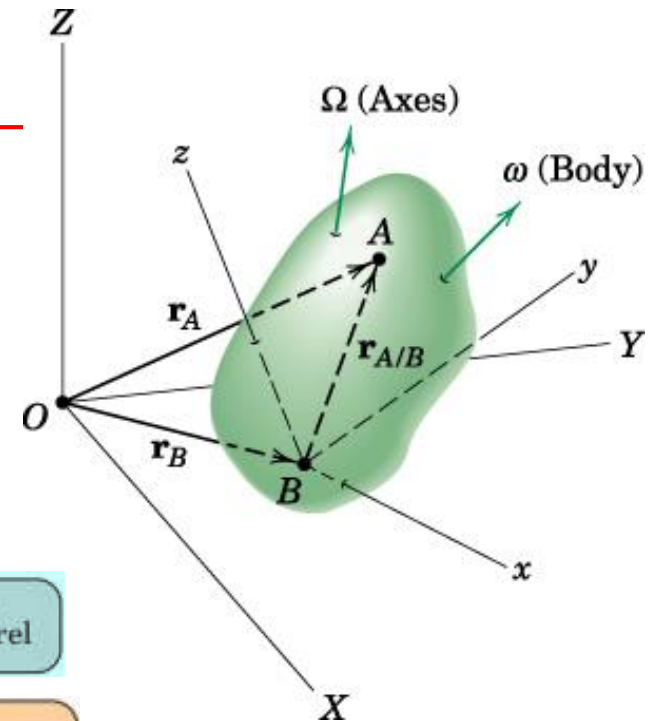
Vel and accln of point A measured rel. to x-y-z by an observer attached to x-y-z

$$\mathbf{v}_{\text{rel}} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k} \quad \text{and} \quad \mathbf{a}_{\text{rel}} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j} + \ddot{z}\mathbf{k}$$

$$\mathbf{a}_A = \mathbf{a}_B + \dot{\boldsymbol{\omega}} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) + 2\boldsymbol{\omega} \times \mathbf{v}_{\text{rel}} + \mathbf{a}_{\text{rel}}$$

: If x-y-z are rigidly attached to the body,

→  $\boldsymbol{\Omega} = \boldsymbol{\omega}$ , and  $\mathbf{v}_{\text{rel}} = \mathbf{a}_{\text{rel}} = 0$  → motion will be the same as for translating reference axes



For different  $\boldsymbol{\Omega}$  and  $\boldsymbol{\omega}$ , the term  $\mathbf{r}_{A/B}$  remains **constant in magnitude** but will **change its direction wrt x-y-z**

# 3-D Kinematics

## Rotating Reference Axes

$$\left(\frac{d\mathbf{V}}{dt}\right)_{XY} = \left(\frac{d\mathbf{V}}{dt}\right)_{xy} + \boldsymbol{\omega} \times \mathbf{V} \quad \text{for any vector}$$

Eqn relating time derivative of a vector  $\mathbf{V}$  measured in fixed  $X$ - $Y$ - $Z$  system and the time derivative of  $\mathbf{V}$  measured relative to rotating  $x$ - $y$ - $z$  system:

$$\left(\frac{d\mathbf{V}}{dt}\right)_{XYZ} = \left(\frac{d\mathbf{V}}{dt}\right)_{xyz} + \boldsymbol{\Omega} \times \mathbf{V}$$

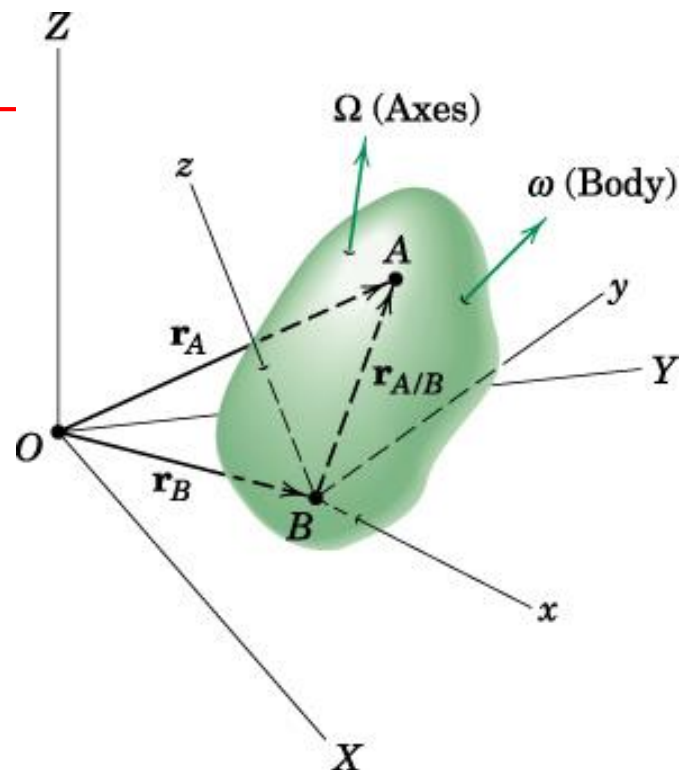
Applying this transformation to the relative posn vector:

$$\mathbf{r}_{A/B} = \mathbf{r}_A - \mathbf{r}_B \quad \rightarrow \quad \left(\frac{d\mathbf{r}_A}{dt}\right)_{XYZ} = \left(\frac{d\mathbf{r}_B}{dt}\right)_{XYZ} + \left(\frac{d\mathbf{r}_{A/B}}{dt}\right)_{xyz} + \boldsymbol{\Omega} \times \mathbf{r}_{A/B}$$

→ The same eqn is obtained

$$\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{\text{rel}} + \boldsymbol{\Omega} \times \mathbf{r}_{A/B}$$

$$\mathbf{a}_A = \mathbf{a}_B + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{A/B} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{A/B}) + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \mathbf{a}_{\text{rel}}$$



# 3-D Kinematics

## Rotating Reference Axes

$$\left(\frac{d\mathbf{V}}{dt}\right)_{XYZ} = \left(\frac{d\mathbf{V}}{dt}\right)_{xyz} + \boldsymbol{\Omega} \times \mathbf{V}$$

This eqn may be recast as the vector operator for general use:

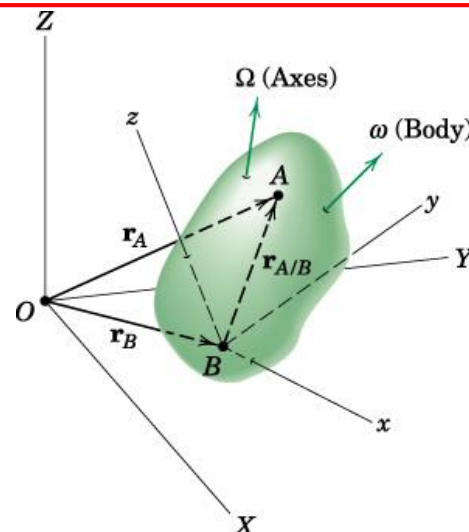
$$\left(\frac{d[\quad]}{dt}\right)_{XYZ} = \left(\frac{d[\quad]}{dt}\right)_{xyz} + \boldsymbol{\Omega} \times [\quad]$$

[ ] stands for any vector  $\mathbf{V}$  expressible both in X-Y-Z and in x-y-z.

Applying the operator to itself → second time derivative.

$$\left(\frac{d^2[\quad]}{dt^2}\right)_{XYZ} = \left(\frac{d^2[\quad]}{dt^2}\right)_{xyz} + \dot{\boldsymbol{\Omega}} \times [\quad] + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times [\quad]) + 2\boldsymbol{\Omega} \times \left(\frac{d[\quad]}{dt}\right)_{xyz}$$

→ The same eqn is for accln



$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\Omega} \times \mathbf{r}_{A/B} + \mathbf{v}_{\text{rel}}$$

$$\mathbf{a}_A = \mathbf{a}_B + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{A/B} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{A/B}) + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \mathbf{a}_{\text{rel}}$$



# Example(1) on Rotating Reference Axes

The motor housing and its bracket rotate about the  $Z$ -axis at the constant rate  $\Omega = 3 \text{ rad/s}$ . The motor shaft and disk have a constant angular velocity of spin  $p = 8 \text{ rad/s}$  with respect to the motor housing in the direction shown. If  $\gamma$  is constant at  $30^\circ$ , determine the velocity and acceleration of point  $A$  at the top of the disk and the angular acceleration  $\alpha$  of the disk.

## Solution

The rotating reference axes  $x$ - $y$ - $z$  are attached to the motor housing.

Unit vectors for  $X$ - $Y$ - $Z$ :  $\mathbf{I}, \mathbf{J}, \mathbf{K}$

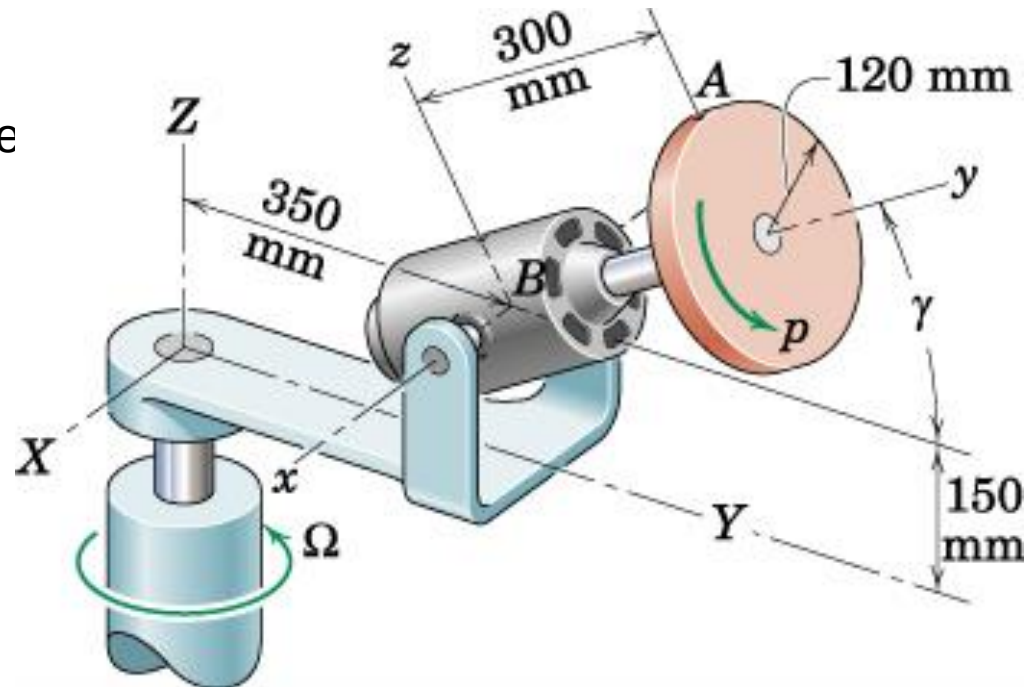
Unit vectors for  $x$ - $y$ - $z$ :  $\mathbf{i}, \mathbf{j}, \mathbf{k}$

Angular vel of  $x$ - $y$ - $z$ :

$$\boldsymbol{\Omega} = \Omega \mathbf{K} = 3 \mathbf{K} \text{ rad/s}$$

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\Omega} \times \mathbf{r}_{A/B} + \mathbf{v}_{\text{rel}}$$

$$\mathbf{a}_A = \mathbf{a}_B + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{A/B} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{A/B}) + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \mathbf{a}_{\text{rel}}$$



# Example (1) on Rotating Reference Axes

Velocity of A:

$$\Omega = \Omega \mathbf{K} = 3\mathbf{K} \text{ rad/s}$$

$$\mathbf{v}_A = \mathbf{v}_B + \Omega \times \mathbf{r}_{A/B} + \mathbf{v}_{\text{rel}}$$

$$\mathbf{v}_B = \Omega \times \mathbf{r}_B = 3\mathbf{K} \times 0.350\mathbf{J} = -1.05\mathbf{I} = -1.05\mathbf{i} \text{ m/s}$$

$$\Omega \times \mathbf{r}_{A/B} = 3\mathbf{K} \times (0.300\mathbf{j} + 0.120\mathbf{k})$$

$$\mathbf{K} \times \mathbf{i} = \mathbf{J} = \mathbf{j} \cos \gamma - \mathbf{k} \sin \gamma$$

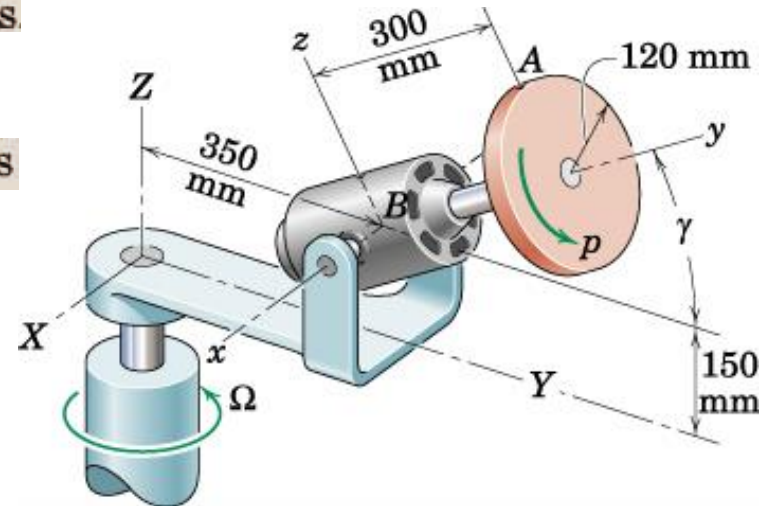
$$\mathbf{K} \times \mathbf{j} = -\mathbf{i} \cos \gamma$$

$$\mathbf{K} \times \mathbf{k} = \mathbf{i} \sin \gamma$$

$$\Omega \times \mathbf{r}_{A/B} = (-0.9 \cos 30^\circ)\mathbf{i} + (0.36 \sin 30^\circ)\mathbf{i} = -0.599\mathbf{i} \text{ m/s}$$

$$\mathbf{v}_{\text{rel}} = \mathbf{p} \times \mathbf{r}_{A/B} = 8\mathbf{j} \times (0.300\mathbf{j} + 0.120\mathbf{k}) = 0.960\mathbf{i} \text{ m/s}$$

$$\rightarrow \mathbf{v}_A = -1.05\mathbf{i} - 0.599\mathbf{i} + 0.960\mathbf{i} = -0.689\mathbf{i} \text{ m/s}$$





# Example (1) on Rotating Reference Axes

## Acceleration of A

$$\mathbf{a}_A = \mathbf{a}_B + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{A/B} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{A/B}) + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \mathbf{a}_{\text{rel}}$$

Here:

$$\begin{aligned}\mathbf{a}_B &= \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_B) = 3\mathbf{K} \times (3\mathbf{K} \times 0.350\mathbf{J}) = -3.15\mathbf{J} \\ &= 3.15(-\mathbf{j} \cos 30^\circ + \mathbf{k} \sin 30^\circ) = -2.73\mathbf{j} + 1.575\mathbf{k} \text{ m/s}^2\end{aligned}$$

$$\dot{\boldsymbol{\Omega}} = \mathbf{0}$$

$$\begin{aligned}\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{A/B}) &= 3\mathbf{K} \times [3\mathbf{K} \times (0.300\mathbf{j} + 0.120\mathbf{k})] \\ &= 3\mathbf{K} \times (-0.599\mathbf{i}) = -1.557\mathbf{j} + 0.899\mathbf{k} \text{ m/s}^2\end{aligned}$$

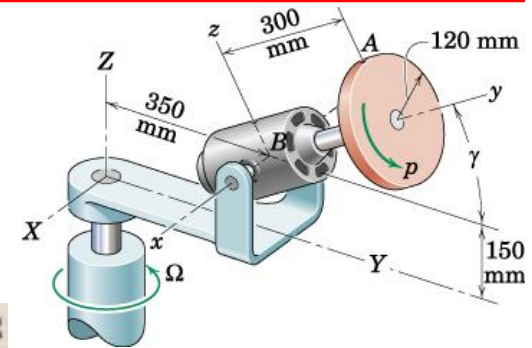
$$\begin{aligned}2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} &= 2(3\mathbf{K}) \times 0.960\mathbf{i} = 5.76\mathbf{J} \\ &= 5.76(\mathbf{j} \cos 30^\circ - \mathbf{k} \sin 30^\circ) = 4.99\mathbf{j} - 2.88\mathbf{k} \text{ m/s}^2\end{aligned}$$

$$\mathbf{a}_{\text{rel}} = \mathbf{p} \times (\mathbf{p} \times \mathbf{r}_{A/B}) = 8\mathbf{j} \times [8\mathbf{j} \times (0.300\mathbf{j} + 0.120\mathbf{k})] = -7.68\mathbf{k} \text{ m/s}^2$$

$$\Rightarrow \mathbf{a}_A = 0.703\mathbf{j} - 8.09\mathbf{k} \text{ m/s}^2 \quad a_A = \sqrt{(0.703)^2 + (8.09)^2} = 8.12 \text{ m/s}^2$$

Angular Accln:

$$\boldsymbol{\alpha} = \dot{\boldsymbol{\omega}} = \boldsymbol{\Omega} \times \boldsymbol{\omega} = 3\mathbf{K} \times (3\mathbf{K} + 8\mathbf{j}) = \mathbf{0} + (-24 \cos 30^\circ)\mathbf{i} = -20.8\mathbf{i} \text{ rad/s}^2$$



# Example on general motion

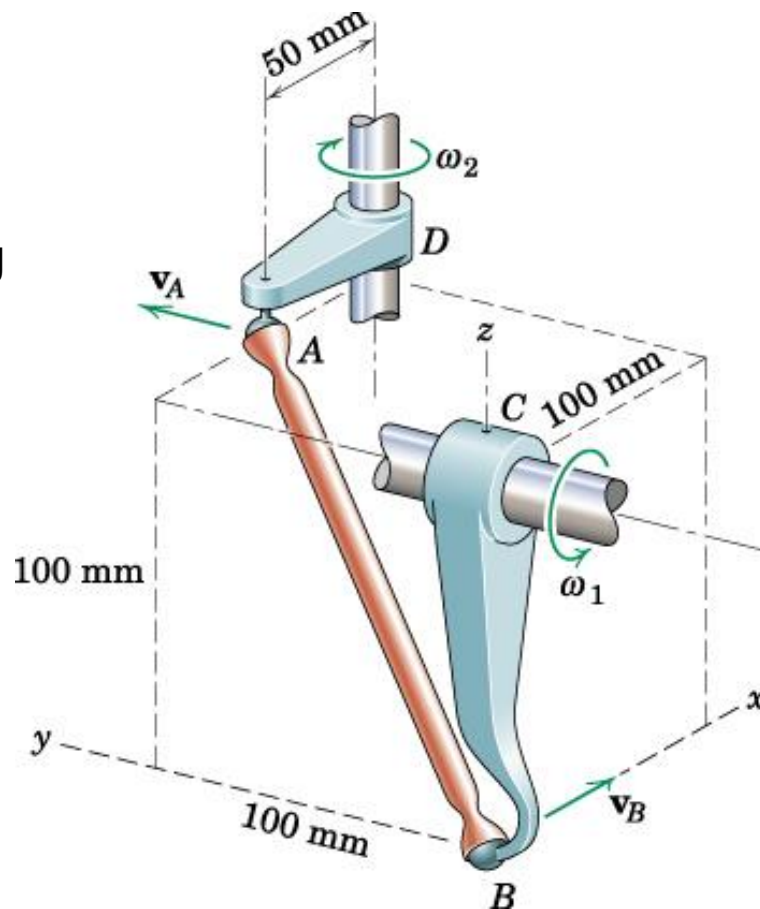
**Example:** Crank  $CB$  rotates @ the horz axis with an ang vel  $\omega_1 = 6 \text{ rad/s}$ , which is constant for a short interval of motion including the position shown. Link  $AB$  has a ball-and-socket fitting on each end and connects crank  $DA$  with  $CB$ . For the instant shown, det the ang vel  $\omega_2$  of crank  $DA$  and ang vel  $\omega_n$  of link  $AB$ . Further determine, the ang accln  $\dot{\omega}_2$  of crank  $DA$  and ang accln  $\dot{\omega}_n$  of link  $AB$ .

## Solution:

Choose point  $B$  as the origin of translating reference axes.

Relative velocity relation:

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\omega}_n \times \mathbf{r}_{A/B}$$



# Example on general motion

Relative velocity relation:

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\omega}_n \times \mathbf{r}_{A/B}$$

$\boldsymbol{\omega}_n$  is the angular vel of link  $AB$  taken normal to  $AB$

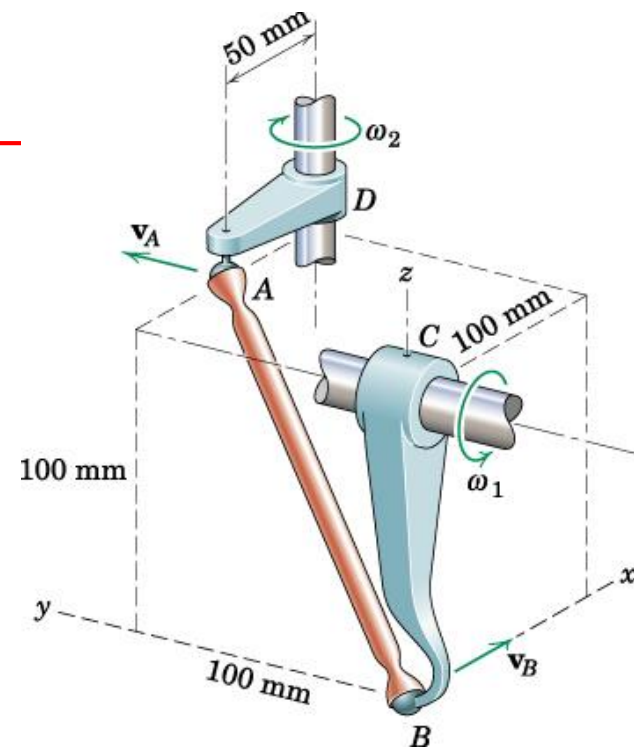
$\therefore$  The angular vel of  $AB$  is taken as normal to  $AB$  since any rotation of the link @ its own axis  $AB$  has no influence on the behaviour of the link.

$\rightarrow \boldsymbol{\omega}_n$  and  $\mathbf{r}_{A/B}$  must be at right angles

$\rightarrow \boldsymbol{\omega}_n \cdot \mathbf{r}_{A/B} = 0$

$\rightarrow$  Additional eqn generated from kinematic requirement

$\mathbf{v}_{A/B}$  is also perpendicular to both  $\boldsymbol{\omega}_n$  and  $\mathbf{r}_{A/B}$   
(since  $\mathbf{v}_A - \mathbf{v}_B = \mathbf{v}_{A/B} = \boldsymbol{\omega}_n \times \mathbf{r}_{A/B}$ )



# Example on general motion

Velocities of A and B:  $v = r \omega$

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\omega}_n \times \mathbf{r}_{A/B}$$

$$\mathbf{v}_A = 50\omega_2 \mathbf{j} \quad \mathbf{v}_B = 100(6)\mathbf{i} = 600\mathbf{i} \text{ mm/s}$$

$$\mathbf{r}_{A/B} = 50\mathbf{i} + 100\mathbf{j} + 100\mathbf{k} \text{ mm} \quad (\text{from } B \text{ to } A)$$

$$\rightarrow 50\omega_2 \mathbf{j} = 600\mathbf{i} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \omega_{n_x} & \omega_{n_y} & \omega_{n_z} \\ 50 & 100 & 100 \end{vmatrix}$$

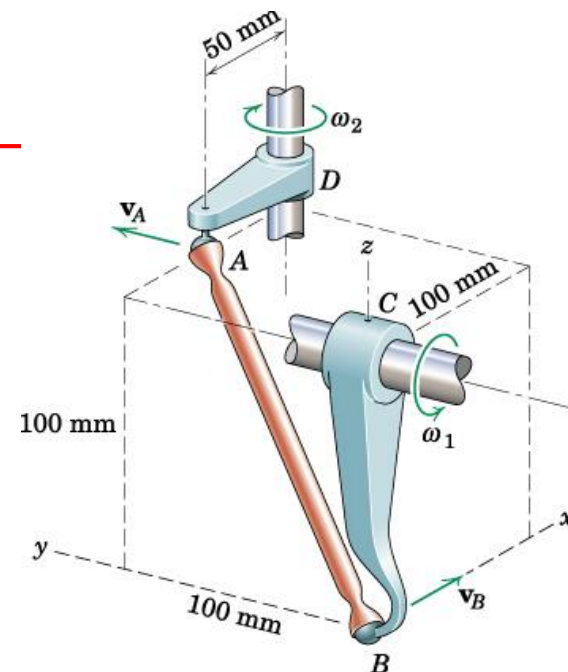
Expanding and equating the coefficients:

$$-6 = \quad + \omega_{n_y} - \omega_{n_z}$$

$$\omega_2 = -2\omega_{n_x} + \omega_{n_z}$$

$$0 = 2\omega_{n_x} - \omega_{n_y}$$

Solving:  $\omega_2 = 6 \text{ rad/s}$



$$\boldsymbol{\omega}_n \cdot \mathbf{r}_{A/B} = 0$$

$$50\omega_{n_x} + 100\omega_{n_y} + 100\omega_{n_z} = 0$$

Substituting in previous two eqns:

$$\omega_{n_x} = -4/3 \text{ rad/s}$$

$$\omega_{n_y} = -8/3 \text{ rad/s}$$

$$\omega_{n_z} = 10/3 \text{ rad/s}$$

$$\boldsymbol{\omega}_n = \frac{2}{3}(-2\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}) \text{ rad/s}$$

$$\omega_n = \frac{2}{3}\sqrt{2^2 + 4^2 + 5^2} = 2\sqrt{5} \text{ rad/s}$$

# Example on general motion

## Angular Acceleration

Linear accln of the links can be found from:

$$\mathbf{a}_A = \mathbf{a}_B + \dot{\boldsymbol{\omega}}_n \times \mathbf{r}_{A/B} + \boldsymbol{\omega}_n \times (\boldsymbol{\omega}_n \times \mathbf{r}_{A/B})$$

$\boldsymbol{\omega}_n$  :: angular velocity of  $AB$  taken normal to  $AB$

$\dot{\boldsymbol{\omega}}_n$  :: the angular accln of link  $AB$

: Comp of ang vel along  $AB$  has no influence on the motion of the cranks  $C$  and  $D$

→ Only the effect of  $\dot{\boldsymbol{\omega}}_n$

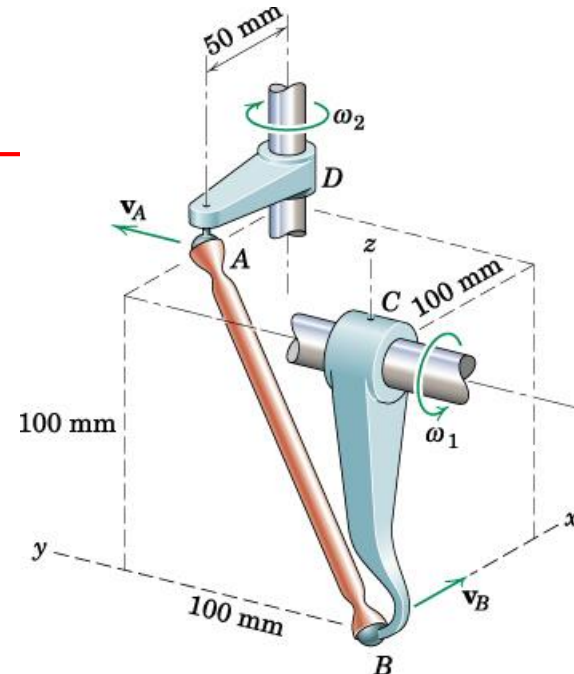
Accln of  $A$  and  $B$  in terms of normal & tangential comp:

$$\mathbf{a}_A = 50\omega_2^2\mathbf{i} + 50\dot{\omega}_2\mathbf{j} = 1800\mathbf{i} + 50\dot{\omega}_2\mathbf{j} \text{ mm/s}^2$$

$$\mathbf{a}_B = 100\omega_1^2\mathbf{k} + (0)\mathbf{i} = 3600\mathbf{k} \text{ mm/s}^2 \quad \dot{\omega}_1 = 0, \text{ since } \omega_1 \text{ is constant}$$

$$\text{Further: } \boldsymbol{\omega}_n \times (\boldsymbol{\omega}_n \times \mathbf{r}_{A/B}) = -\omega_n^2 \mathbf{r}_{A/B} = -20(50\mathbf{i} + 100\mathbf{j} + 100\mathbf{k}) \text{ mm/s}^2$$

$$\mathbf{P} \times (\mathbf{Q} \times \mathbf{R}) = (\mathbf{P} \cdot \mathbf{R})\mathbf{Q} - (\mathbf{P} \cdot \mathbf{Q})\mathbf{R} \text{ (first term will be zero because } \boldsymbol{\omega}_n \text{ is normal to } \mathbf{r}_{A/B})$$



$$a_n = |\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})| = b\omega^2$$

$$a_t = |\dot{\boldsymbol{\omega}} \times \mathbf{r}| = b\alpha$$



# Example on general motion

$$\mathbf{a}_A = \mathbf{a}_B + \dot{\boldsymbol{\omega}}_n \times \mathbf{r}_{A/B} + \boldsymbol{\omega}_n \times (\boldsymbol{\omega}_n \times \mathbf{r}_{A/B})$$

$$\begin{aligned} \dot{\boldsymbol{\omega}}_n \times \mathbf{r}_{A/B} = & (100\dot{\omega}_{n_y} - 100\dot{\omega}_{n_z})\mathbf{i} \\ & + (50\dot{\omega}_{n_z} - 100\dot{\omega}_{n_x})\mathbf{j} + (100\dot{\omega}_{n_x} - 50\dot{\omega}_{n_y})\mathbf{k} \end{aligned}$$

Substituting into rel. accln. eqn and equating respective coefficients:

$$\begin{aligned} 28 &= \dot{\omega}_{n_y} - \dot{\omega}_{n_z} \\ \dot{\omega}_2 + 40 &= -2\dot{\omega}_{n_x} + \dot{\omega}_{n_z} \Rightarrow \dot{\omega}_2 = -36 \text{ rad/s}^2 \\ -32 &= 2\dot{\omega}_{n_x} - \dot{\omega}_{n_y} \end{aligned}$$

The vector  $\boldsymbol{\omega}_n$  is normal to  $\mathbf{r}_{A/B}$  (but is not normal to  $\mathbf{v}_{A/B}$ ). The component of  $\boldsymbol{\omega}_n$  which is not normal to  $\mathbf{v}_{A/B}$  gives rise to the change in direction of  $\mathbf{v}_{A/B}$ .

$$[\dot{\boldsymbol{\omega}}_n \cdot \mathbf{r}_{A/B} = 0] \quad 2\dot{\omega}_{n_x} + 4\dot{\omega}_{n_y} + 4\dot{\omega}_{n_z} = 0$$

Substituting in previous two eqns:  $\dot{\omega}_{n_x} = -8 \text{ rad/s}^2$   $\dot{\omega}_{n_y} = 16 \text{ rad/s}^2$   $\dot{\omega}_{n_z} = -12 \text{ rad/s}^2$

$$\dot{\boldsymbol{\omega}}_n = 4(-2\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}) \text{ rad/s}^2 \quad |\dot{\boldsymbol{\omega}}_n| = 4\sqrt{2^2 + 4^2 + 3^2} = 4\sqrt{29} \text{ rad/s}^2$$

